



Abstract

- Framework for remote sensing segmentation, captioning and synthetic data generation
- 100,240 real and 300,000 synthetic images combined with segmentation maps and captions
- A total of 2,001,200 captions generated by different Vision and Language models^{1,2}
- Synthetic captions match human-level CLIPScore³
- High volume and less redundant than alternatives
- Image segmentation model training benchmark
- Competitive performance with only synthetic data
- Improved results with synthetic data augmentation

Problem Definition

Redundant captions

Caption (1-5): khaki beach is between green trees and blue ocean →



Repetitive captions

← This is a dense forest with lots of dark green trees →

Existing remote sensing caption datasets^{4,5,6}:

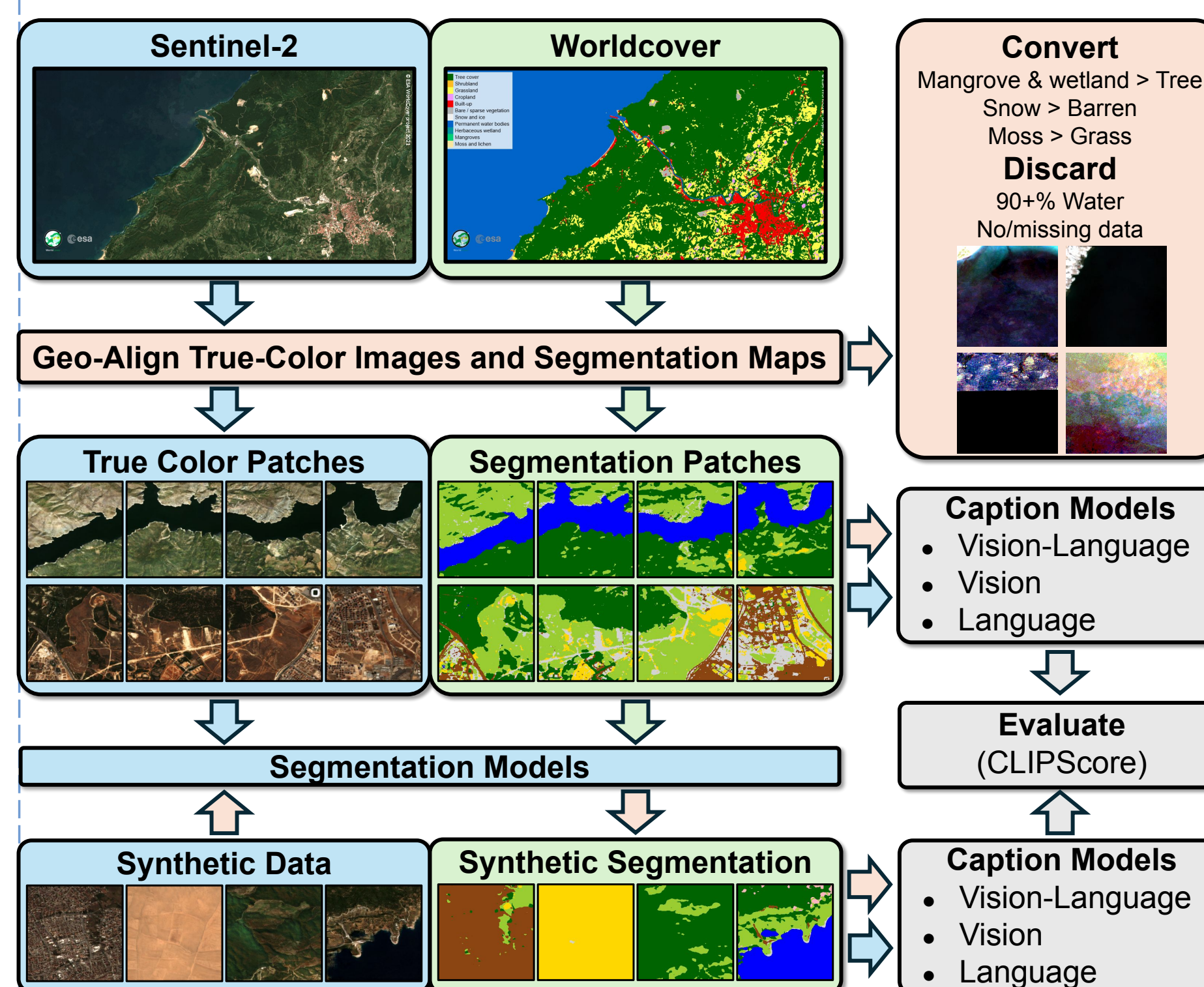
- × Limited volume (require labor-intensive captioning)
- × Often contain repetitive captions
- × Prone to human-error

Our approach:

- ✓ Larger than existing datasets and scalable volume
- ✓ Varied captions with low redundancy
- ✓ Consistent and aligns with human assessment
- ✓ Synthetic data improves segmentation performance

Methods and Materials

- Data collection (Sentinel-2⁷ RGB and Worldcover⁸)
- Synthetic sample generation^{9,10,11}
- VLM and LLM assisted automated captioning

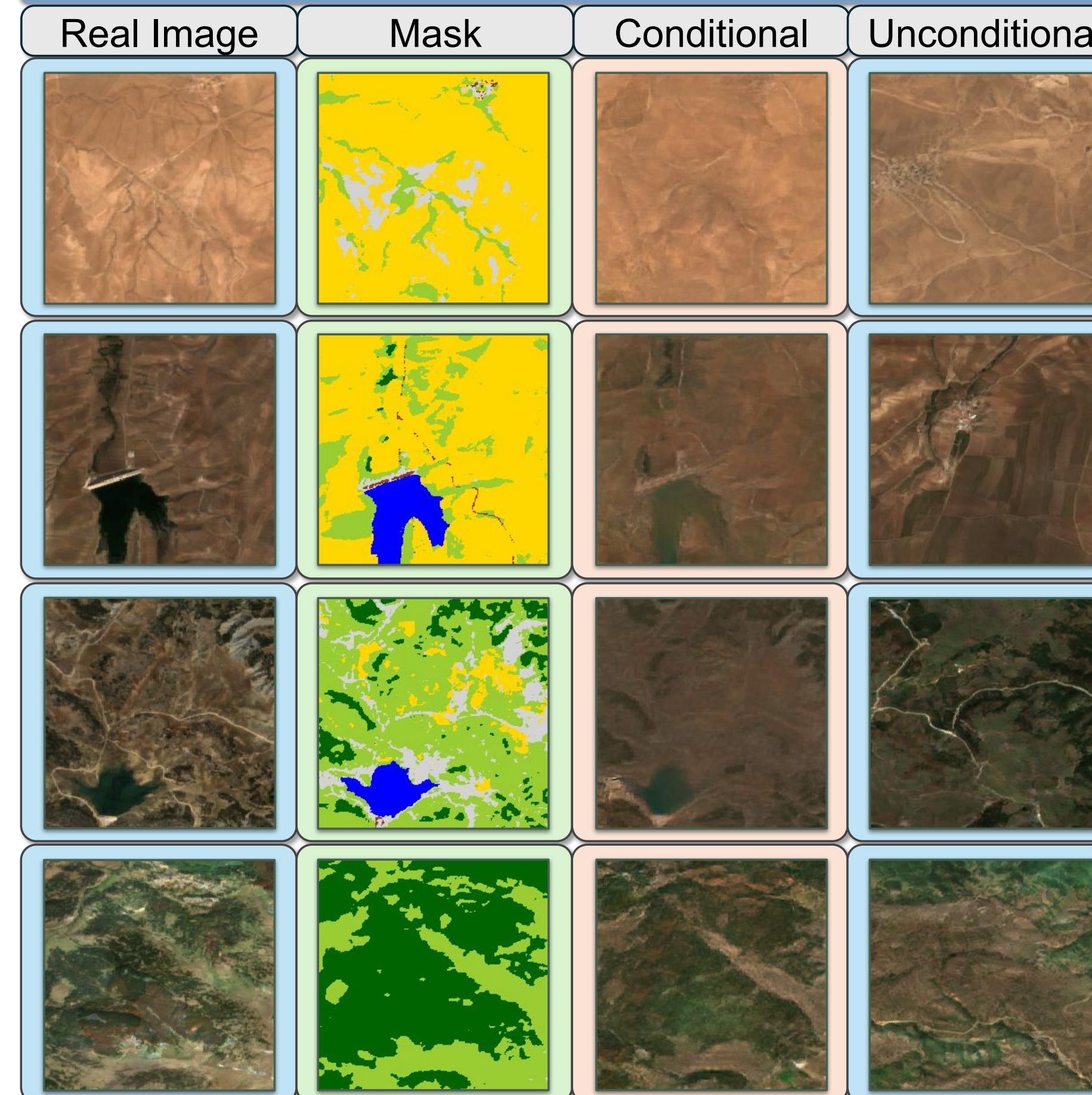


Data collection, synthetic sample generation and captioning framework

Dataset	Volume	Captions	Redundancy↓	CLIPScore↑
NWPU ⁴	31,500	157,500	72.65%	30.25
RSICD ⁵	10,921	54,605	67.02%	29.11
UCMC ⁶	2,100	10,500	80.88%	30.18
ARAS400k	400,240	2,001,200	12.85%	29.66
-Real	100,240	501,200	8.01%	29.89
-Synth	300,000	1,500,000	13.06%	29.58

Volume, redundancy and CLIPScore of our dataset and literature alternatives

Results

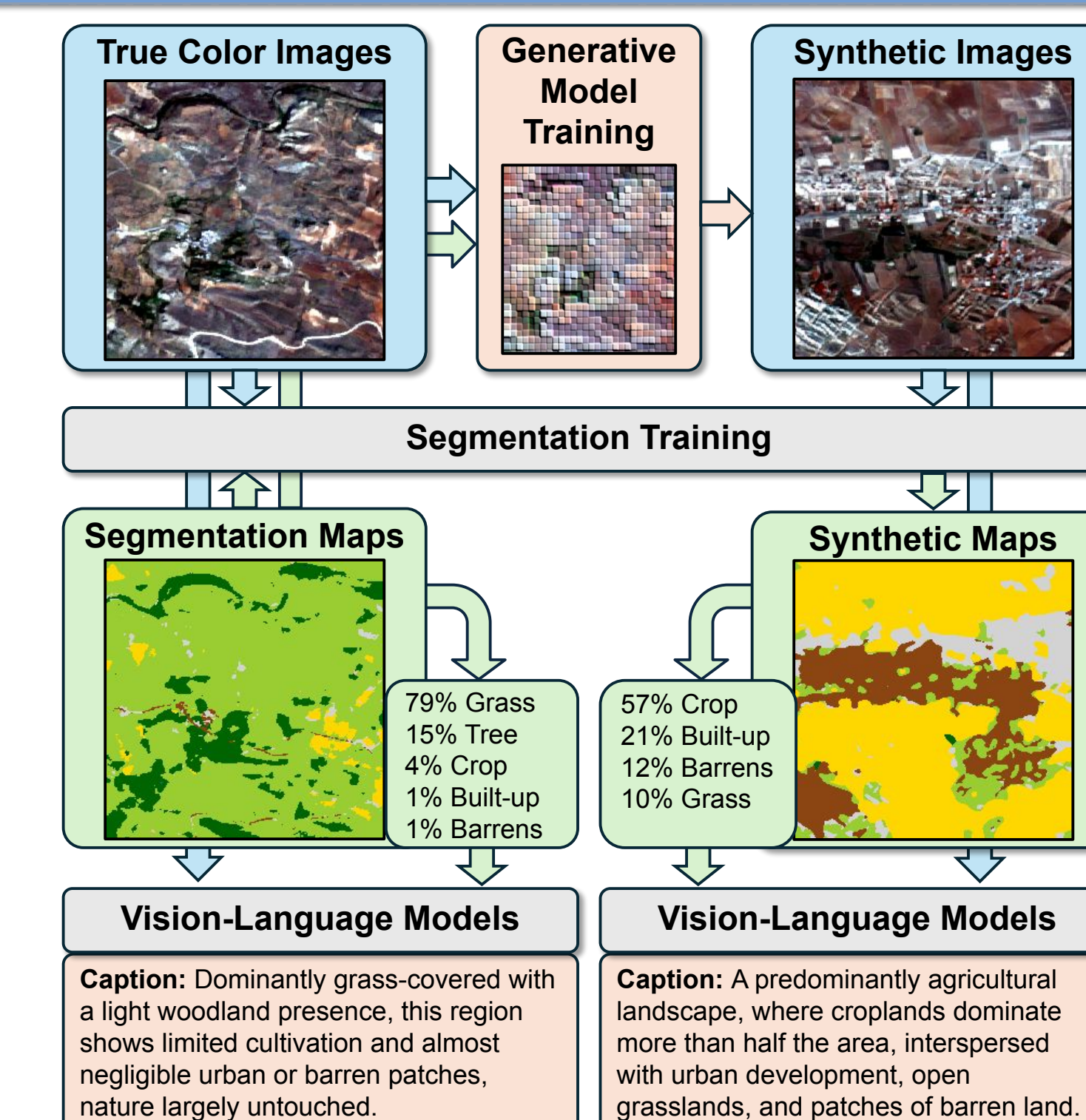


Benchmark Results

Model	Real Data				Synthetic Data				Real + Synthetic Data			
	F1	IoU	Prec.	Rec.	F1	IoU	Prec.	Rec.	F1	IoU	Prec.	Rec.
Unet	76.37	64.49	77.05	75.86	74.63	62.31	77.60	72.23	77.40	65.58	79.37	75.64
Unet++	76.38	64.36	76.50	76.39	74.94	62.73	77.81	72.56	78.23	66.67	80.05	76.65
PAN	76.20	64.10	75.29	77.60	74.65	62.42	77.70	72.22	77.21	65.48	79.94	75.02
DeepLab	76.02	63.91	75.59	76.77	74.63	62.39	77.69	72.15	77.62	65.94	80.00	75.65
Segformer	77.09	65.26	78.10	76.33	75.10	62.90	77.65	72.95	77.80	66.14	79.98	75.98
FPN	76.85	64.92	77.07	76.76	74.81	62.59	77.99	72.26	77.71	66.04	80.16	75.66
Mean	76.49	64.51	76.60	76.62	74.79	62.56	77.74	72.40	77.66	65.98	79.92	75.77

Segmentation model performance comparison using ARAS400k-subsets

Discussion



Generative data augmentation pipeline starting from real true color images

Conclusions

- Open source dataset generation pipeline
- Publicly available dataset and results
- Clean and complete image segmentation dataset
- VLM driven automated large-scale captioning
- Results comparable with human captions
- Publicly available synthetic extension data
- Improved segmentation downstream performance

Future Works

- Super-resolution applications at scale
- Improve structure-preserving conditional generation
- Extend to other domains (e.g. medical imaging)

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Docker Image



Dataset



Code Base



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